

PROBABILITY — MCQ ASSESSMENT

PART A — BASIC PROBABILITY

1. The probability of an event can have a value between:

- A. -1 and 1
- B. 0 and 1
- C. 1 and 100
- D. $-\infty$ and $+\infty$

2. If the probability of an event is 0, the event is:

- A. Certain
- B. Likely
- C. Impossible
- D. Equally likely

3. If the probability of an event is 1, the event is:

- A. Impossible
- B. Certain
- C. Unlikely
- D. Random

4. A fair coin is tossed once. What is the probability of getting Heads?

- A. 0
- B. 0.25
- C. 0.5
- D. 1

5. A fair six-sided die is rolled once. What is the probability of getting a 4?

- A. $\frac{1}{2}$
- B. $\frac{1}{4}$
- C. $\frac{1}{6}$
- D. $\frac{4}{6}$

6. A fair die is rolled once. What is the probability of getting an even number?

- A. $\frac{1}{6}$
- B. $\frac{1}{3}$
- C. $\frac{1}{2}$
- D. $\frac{2}{3}$

7. If $P(A) = 0.3$, what is $P(\text{not } A)$?

- A. 0.3
- B. 0.5
- C. 0.7
- D. 1.3

8. Which of the following represents the complement of event A?

- A. $A + B$
- B. $A \cap B$

- C. $A \cup B$
- D. A'

PART B — EVENTS

9. A sample space represents:

- A. Only successful outcomes
- B. All possible outcomes of an experiment
- C. Only failed outcomes
- D. The probability of an event

10. When a die is rolled once, the sample space is:

- A. $\{0, 1, 2, 3, 4, 5\}$
- B. $\{1, 2, 3, 4, 5, 6\}$
- C. $\{2, 4, 6\}$
- D. $\{1, 3, 5\}$

11. If A represents getting an even number when rolling a die, which set represents A?

- A. $\{1, 3, 5\}$
- B. $\{2, 4, 6\}$
- C. $\{1, 2, 3\}$
- D. $\{4, 5, 6\}$

12. Two events are mutually exclusive when:

- A. They always occur together
- B. They cannot occur at the same time
- C. They have equal probability
- D. They have equal outcomes

13. Which pair of events is mutually exclusive when rolling a die?

- A. Getting an even number and getting a 4
- B. Getting a number greater than 3 and getting a 6
- C. Getting an odd number and getting an even number
- D. Getting a number less than 5 and getting a 3

PART C — ADDITION RULE

14. If two events A and B are mutually exclusive, which expression gives $P(A \text{ or } B)$?

- A. $P(A) \times P(B)$
- B. $P(A) + P(B)$
- C. $P(A) - P(B)$
- D. $P(A) / P(B)$

15. If $P(A) = 0.4$ and $P(B) = 0.3$, and A and B are mutually exclusive, what is $P(A \text{ or } B)$?

- A. 0.12
- B. 0.1
- C. 0.7
- D. 1.2

16. If two events can occur at the same time, simply adding their probabilities may:

- A. Always give the correct answer
- B. Double-count their overlap
- C. Make the probability zero
- D. Make the events independent

17. Suppose:

$$P(A) = 0.5$$

$$P(B) = 0.4$$

$$P(A \text{ and } B) = 0.2$$

What is $P(A \text{ or } B)$?

- A. 0.2
- B. 0.7
- C. 0.9
- D. 1.1

PART D — CONDITIONAL PROBABILITY

18. Conditional probability answers which question?

- A. What is the probability of A given that B has occurred?
- B. What is the probability of A and B never occurring?
- C. What is the probability of all possible events?
- D. What is the average probability?

19. The notation $P(A | B)$ means:

- A. Probability of A and B
- B. Probability of A given B
- C. Probability of B given A
- D. Probability of neither A nor B

20. If $P(A \text{ and } B) = 0.2$ and $P(B) = 0.5$, what is $P(A | B)$?

- A. 0.1
- B. 0.2
- C. 0.4
- D. 0.7

21. A medical test is performed only on patients who have already been identified as high-risk. The probability of a positive result in this group is an example of:

- A. Conditional probability
- B. Independent probability
- C. Complementary probability
- D. Uniform probability

22. If knowing that event B occurred changes the probability of event A, then A and B are generally:

- A. Independent
- B. Dependent
- C. Mutually exclusive
- D. Identical

PART E — INDEPENDENT EVENTS

23. Two events are independent when:

- A. They cannot occur together
- B. The occurrence of one does not affect the probability of the other
- C. They always occur together
- D. Their probabilities are equal

24. A fair coin is tossed twice. Is the result of the second toss affected by the result of the first toss?

- A. Yes, always
- B. No
- C. Only when the first toss is Heads
- D. Only when the first toss is Tails

25. If A and B are independent, which expression gives $P(A \text{ and } B)$?

- A. $P(A) + P(B)$
- B. $P(A) - P(B)$
- C. $P(A) \times P(B)$
- D. $P(A) / P(B)$

26. If $P(A) = 0.5$ and $P(B) = 0.2$, and A and B are independent, what is $P(A \text{ and } B)$?

- A. 0.1
- B. 0.3
- C. 0.7
- D. 1.0

27. Which pair of events is most likely to be independent?

- A. Drawing two cards from a deck without replacement
- B. Tossing a coin twice
- C. Selecting two students without replacement
- D. Taking two balls from a bag without replacement

PART F — COUNTING AND COMBINATORICS

28. A fair die is rolled twice. How many possible ordered outcomes are there?

- A. 6
- B. 12
- C. 36
- D. 42

29. A coin is tossed three times. How many possible outcomes are there?

- A. 3
- B. 6
- C. 8
- D. 9

30. If there are 4 shirts and 3 pairs of trousers, how many different shirt-trouser combinations are possible?

- A. 7
- B. 12**
- C. 16
- D. 24

31. Which concept is used when the order of selection matters?

- A. Combination
- B. Permutation**
- C. Mean
- D. Variance

32. Which concept is used when the order of selection does NOT matter?

- A. Permutation
- B. Combination**
- C. Conditional probability
- D. Independence

33. In how many ways can 3 different students be arranged in a row?

- A. 3
- B. 6**
- C. 9
- D. 12

PART G — RANDOM VARIABLES

34. A random variable is:

- A. A variable whose value is determined by the outcome of a random experiment**
- B. A variable that must always be random in a computer program
- C. A variable that can only contain integers
- D. A variable with probability 1

35. Which of the following is an example of a discrete random variable?

- A. Height of a person
- B. Temperature
- C. Number of defective products**
- D. Weight of a person

36. Which of the following is an example of a continuous random variable?

- A. Number of students
- B. Number of defective items
- C. Number of cars
- D. Temperature**

37. The sum of probabilities of all possible outcomes of a discrete random variable should be:

- A. 0
- B. 0.5
- C. 1**
- D. Greater than 1

PART H — EXPECTED VALUE

38. Expected value can be interpreted as:

- A. The most frequent outcome in every experiment
- B. The long-run weighted average outcome
- C. The maximum possible outcome
- D. The minimum possible outcome

39. A random variable X has:

$X = 0$ with probability 0.5
 $X = 10$ with probability 0.5

What is its expected value?

- A. 2.5
- B. 5
- C. 10
- D. 20

40. A game gives ₹100 with probability 0.2 and ₹0 with probability 0.8. What is the expected value?

- A. ₹10
- B. ₹20
- C. ₹50
- D. ₹80

PART I — BAYESIAN THINKING

41. Bayes' theorem is primarily concerned with:

- A. Updating probabilities based on new evidence
- B. Calculating averages
- C. Finding matrix rank
- D. Calculating variance

42. Suppose a medical test is positive. You want to know the probability that the patient actually has the disease. Which type of probability is most relevant?

- A. $P(\text{Test Positive})$
- B. $P(\text{Disease})$
- C. $P(\text{Disease} \mid \text{Test Positive})$
- D. $P(\text{Test Positive} \mid \text{No Disease})$

43. Why can a highly accurate medical test still produce many false positives?

- A. The disease may be very common
- B. The disease may be very rare, affecting the base rate
- C. Probability cannot be used for medical testing
- D. Accuracy and probability are always unrelated

44. In Bayesian reasoning, prior probability represents:

- A. Probability after observing evidence
- B. Initial belief before considering new evidence
- C. Probability of an impossible event
- D. Probability of two independent events

PART J — APPLICATION-BASED QUESTIONS

45. A fraud detection system flags 5% of all transactions. If a transaction is randomly selected, what does 5% represent?

- A. Conditional probability
- B. Prior probability
- C. Expected value
- D. Variance

46. A dataset contains 1,000 transactions, of which 50 are fraudulent. What is the probability that a randomly selected transaction is fraudulent?

- A. 0.005
- B. 0.05
- C. 0.5
- D. 5

47. A spam filter identifies an email as spam. You want to know the probability that it actually is spam. Which probability is most relevant?

- A. $P(\text{Spam})$
- B. $P(\text{Flagged})$
- C. $P(\text{Spam} \mid \text{Flagged})$
- D. $P(\text{Flagged} \mid \text{Not Spam})$

48. A classifier predicts that 90% of emails are legitimate. Does this necessarily mean that 90% of its predictions are correct?

- A. Yes, always
- B. No
- C. Only if the dataset is large
- D. Only if the classes are balanced

49. If an event has probability 0.8, which statement is correct?

- A. It will occur exactly 80% of the time in every small sample
- B. It is guaranteed to occur
- C. Over many repetitions, its relative frequency is expected to approach 0.8
- D. It cannot fail

50. A model predicts whether a patient has a disease. The disease occurs in only 1% of the population. Why is the 1% figure important when interpreting a positive prediction?

- A. It represents the base rate
- B. It represents the model accuracy
- C. It represents the model's variance
- D. It represents the sample size